Transient-Impulsivity Features Trained on Slider Units Show Reduced Performance Drop When Applied to Valve Units

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Abstract. Anomaly detection in industrial machinery relies on acoustic monitoring, but detectors trained on one machine unit often fail to generalize to others. This study examines short factory-floor recordings of solenoid valves and slide rails, each operating normally or with an injected mechanical fault, using one-class anomaly detectors trained separately on different feature representations. When trained on slider units, detectors built from transient-impulsivity statistics such as kurtosis and crest factor generalized to other slider units and to valve units with a substantially smaller performance drop than detectors relying on conventional spectral or MFCC energy features. This advantage was specific to models trained on slider data, indicating a machine-type-dependent interaction rather than a universal superiority of impulsivity-based features. Detectors trained on valve units did not show the same generalization benefit, underscoring that feature effectiveness depends on the source machine's fault dynamics. These impulsivity statistics capture transient, shock-like signatures that appear to transfer across mechanically related units more robustly than smooth spectral energy patterns. The findings suggest that feature choice for cross-machine anomaly detection should be guided by the underlying mechanical fault behavior rather than assumed to be universally optimal. This has practical significance for designing scalable, transferable condition-monitoring systems in heterogeneous industrial environments.

1 Introduction

Solenoid valves and slide rails are essential actuation and guidance components in industrial machinery, and their unnoticed degradation can precipitate unplanned downtime, costly repairs, or cascading failures across a production line. Solenoid valves regulate fluid and gas flow in systems ranging from pneumatic actuators to safety shutoffs, so an undetected fault can compromise process control or safety interlocks S et al. [2025], Anonymous [2018]. Slide rails, which provide the mechanical guidance for moving assemblies such as pantograph slide plates, are similarly critical, and their wear or misalignment has been linked to broader mechanical failure in transport and manufacturing contexts Liu [2016], Bruzelius and Mba [2004]. Because these components often operate embedded within noisy factory environments, early and reliable identification of abnormal mechanical behavior, well before catastrophic failure, is essential for maintaining operational continuity, reducing maintenance cost, and ensuring workplace safety.

Condition monitoring research has largely converged on impulsivity-sensitive statistics, such as kurtosis, spectral kurtosis, and envelope demodulation, as the preferred descriptors for impact-like degradation in rotating and reciprocating machinery, with demonstrated success in bearing fault diagnosis Tian et al. [2013], Wang et al. [2017], CHEN [2018], Bai and Xi [2018]. These methods exploit the transient, high-frequency energy bursts that faults produce, and related work on valve core displacement and gate valve anomaly scoring has shown that mechanical irregularities in actuation components leave similarly impulsive acoustic or vibrational signatures Zhao et al. [2018], Luo et al. [2021]. Parallel efforts in acoustic anomaly detection have instead relied on spectral or learned embedding representations, including MFCC-based and image-transfer approaches, trained in a one-class manner on normal sounds Müller et al. [2021], Almudévar et al. [2023]. Most reported evaluations pair matched training and test units of a single machine, leaving open whether transient-impulsivity statistics or conventional mean-spectral and MFCC energy descriptors better

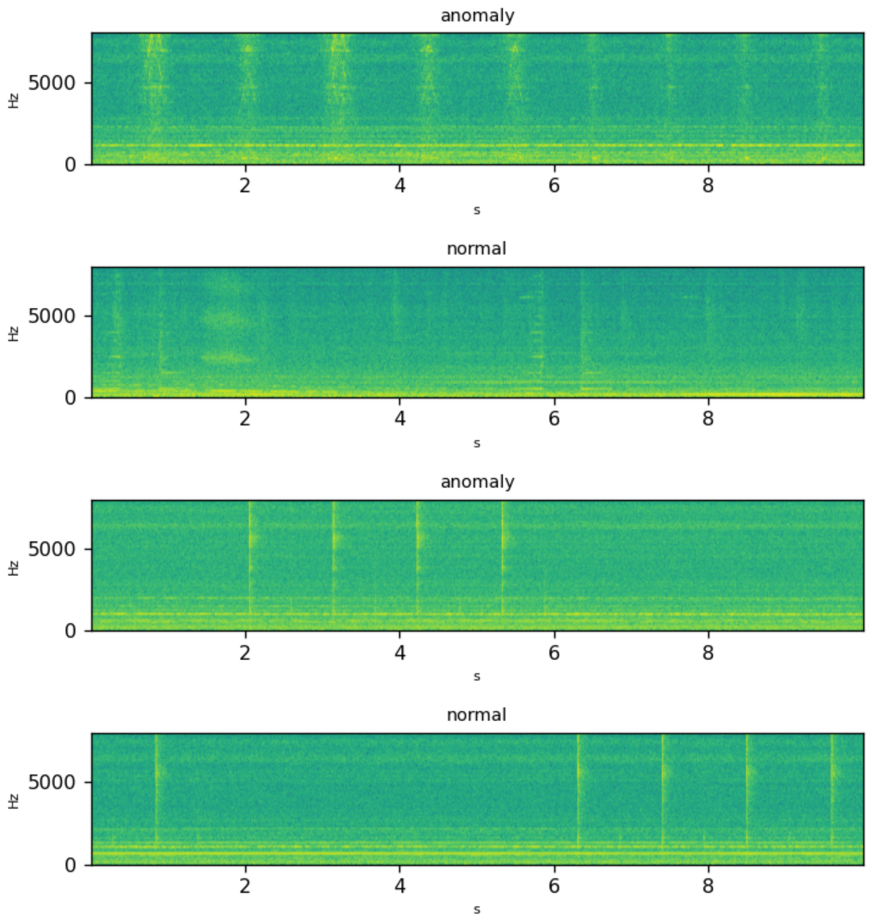

Figure 1. Log-frequency spectrograms of representative single-channel recordings from the dataset (class label above each panel).

support cross-unit and cross-type generalization in a strictly one-class setting, where detection must be learned from normal sounds alone, without access to labeled faults.

We address this gap using short, single-channel recordings of solenoid valve and slide rail units, each either operating normally or exhibiting an injected mechanical fault against a factory-noise background. Specifically, we trained one-class anomaly detectors solely on normal-condition sounds from a single unit and evaluated whether scoring based on transient-impulsivity statistics, namely time-domain kurtosis, crest factor, spectral kurtosis, and envelope sparsity, generalized more robustly to unseen units and to the other machine type than scoring based on conventional mean-spectral and MFCC energy features.

2 Results

2.1 Main Result: Sign-Reversing Interaction

Figure 1 shows representative examples of the raw audio clips (shown as spectrograms) analysed here. The central result of this study is a sign-reversing interaction between machine type and feature class, directly visible in Figure 2 and confirmed in the full cross-group AUC matrices of Figure 5. When one-class detectors are trained on SLIDER units, the mean generalization gap (spectral gap minus impulsivity gap)

Table 1. Key quantitative results across analyses (AUC unless noted).

Analysis	Metric	Values	Note
Type interaction	Mahal. mean-diff	slider +0.219	p=2.4e-4, FDR=1.4e-3
		valve −0.125	p=0.070, FDR=0.114
Baseline impulsivity	kurtosis (mean)	slider 0.33 / valve 17.91	sd 0.69/10.07
	crest factor	slider 5.53 / valve 20.91	sd 1.35/4.26
Ablation (within/crossID/crosstype)	kurtosis	0.689/0.518/0.457	single-feature AUC
	crest_factor	0.751/0.598/0.485	
	spec_kurtosis	0.613/0.564/0.545	
	transient_rate	0.652/–/–	
Domain shift	impulsivity	0.751 (within) / 0.550 (cross)	chance=0.5
	spectral	0.809 (within) / 0.560 (cross)	
Leakage check	impulsivity	0.562 (pooled) / 0.751 (group-disj.)	
	spectral	0.706 (pooled) / 0.809 (group-disj.)	

is positive under both detectors: 0.219 for the robust-covariance Mahalanobis detector (FDR-corrected p equal to 0.0014) and 0.129 for Isolation Forest (FDR-corrected p equal to 0.035), indicating that transient-impulsivity statistics such as kurtosis and crest factor carry information consistent with anomalous behavior with substantially less cross-unit and cross-type performance loss than spectral or MFCC energy features. When the same detectors are instead trained on VALVE units, this advantage inverts: the mean gap becomes negative, -0.125 for Mahalanobis and -0.268 for Isolation Forest (FDR-corrected p equal to 0.0014), showing that impulsivity features generalize worse than spectral features in that regime. Within-domain performance itself is high and comparable for both feature families (impulsivity AUC 0.751, spectral AUC 0.809), so the reversal reflects differential generalization rather than a baseline detection deficit. Figure 3 contextualizes the asymmetry by contrasting the distribution of kurtosis and crest factor for NORMAL clips across the two machine types, foreshadowing the boundary condition developed in the next paragraph. These slider-trained and valve-trained estimates, each derived from the same two independent one-class detectors and surviving false-discovery-rate correction, constitute the paper's core evidence that the impulsivity-versus-spectral generalization gap is machine-type-dependent rather than a fixed property of either feature family.

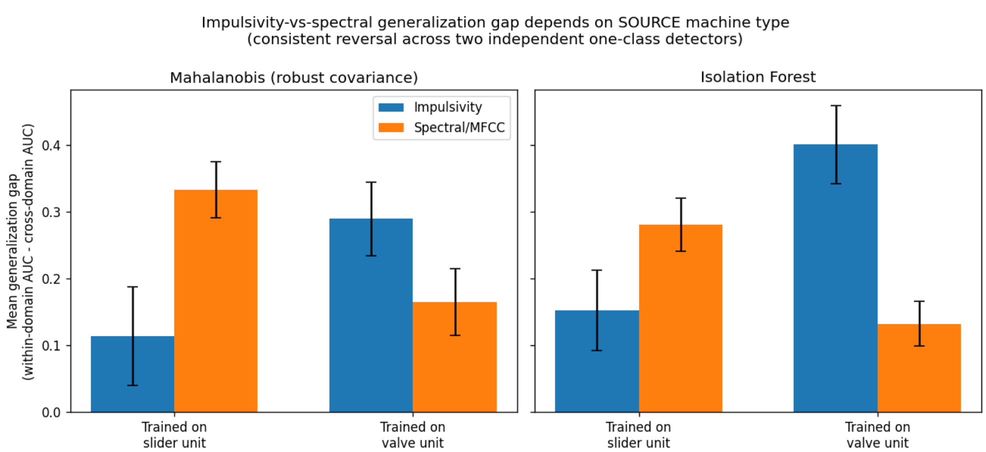

Figure 2. Mean generalization gap (within-domain AUC minus cross-domain AUC) for impulsivity vs spectral/MFCC features, split by whether the one-class model was trained on a slider or a valve unit, shown for two independent detectors (Mahalanobis robust-covariance and Isolation Forest). The advantage of impulsivity features flips sign between machine types.

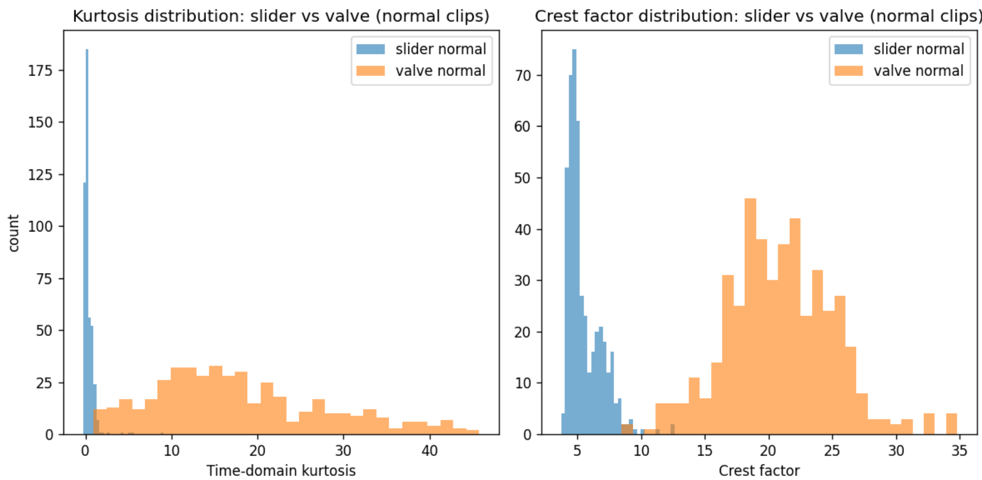

Figure 3. Distribution of time-domain kurtosis and crest factor for NORMAL clips, slider vs valve. Valve's normal operating sound is intrinsically far more impulsive (periodic seating clicks), explaining why impulsivity statistics are not anomaly-specific, and hence don't generalize, for valve-trained models.

2.2 Robustness and Sensitivity Analyses

Robustness and sensitivity checks bound the strength of the machine-type-interaction claim while leaving its core intact. Decomposing the impulsivity set into its five individual components, as shown in Figure 4, indicates that kurtosis and crest factor alone carry the generalization advantage: these two peak-amplitude statistics account for nearly all of the within-domain AUC and gap reduction, whereas spectral kurtosis, envelope gini, and transient rate contribute markedly smaller within-domain AUC and only marginal gap changes. Across the two one-class detectors, the slider-trained effect varies in size: the Mahalanobis mean difference of 0.219 (p false-discovery-rate 0.0014) is roughly 70 percent larger than the Isolation Forest mean difference of 0.129 (p false-discovery-rate 0.035), consistent in direction but variable in magnitude. The valve-trained reversal is more consistent in size across detectors, with mean differences of negative 0.125 and negative 0.268, and the cross-type condition produces the most extreme and detector-concordant reversal (Mahalanobis negative 0.325, Isolation Forest negative 0.430, both surviving false-discovery-rate correction at 0.00027), the sign flip is the most dependable feature of the interaction, even where its magnitude is not.

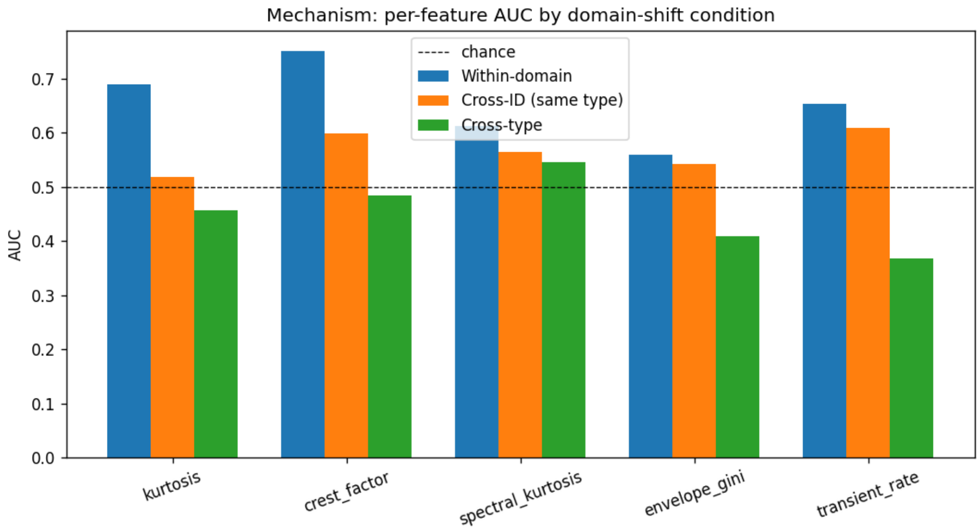

Figure 4. Within-domain, cross-ID, and cross-type AUC for each of the 5 individual impulsivity features (1-D anomaly score = |z-score| vs train-normal mean/std).

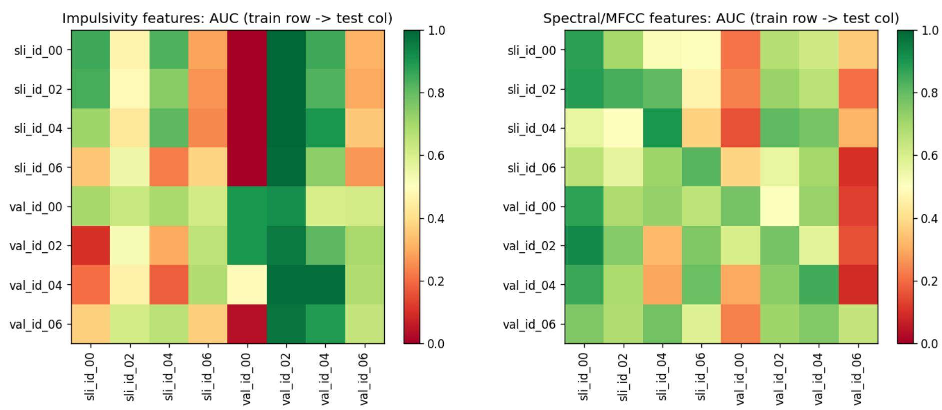

Figure 5. Full 8x8 AUC matrices (train group row -> test group column) for impulsivity (left) and spectral/MFCC (right) one-class Mahalanobis detectors, showing the underlying data behind all gap computations.

3 Discussion

The pattern of results is consistent with a machine-type-dependent organization of fault-relevant information rather than a general property of transient-impulsivity statistics. When the one-class detector is trained on SLIDER units, kurtosis and crest factor carry information that remains target-*like* across other slider units and across valve units under this detector, while the conventional spectral and MFCC energy features suffer a substantially larger generalization gap under the same training regime. This asymmetry suggests that slider faults manifest primarily as transient, impulsive departures from the normal operating envelope, departures that a covariance-based or density-based one-class boundary can localize efficiently in the impulsivity feature space regardless of the recording unit or machine type. The reversal observed for valve-trained detectors, where impulsivity generalizes worse than spectral features, indicates that this advantage is not intrinsic to the statistics themselves but depends on how each machine class expresses its fault signature, with valve faults evidently better captured by spectral or energy-based descriptors. The sign-reversing interaction therefore supports a qualitative claim: impulsivity features are not universally superior anomaly descriptors but instead align preferentially with the acoustic character of slider degradation, making their generalization advantage conditional on machine type rather than a fixed property of one-class detection pipelines.

3.1 Relation to Prior Practice

This pattern extends conventional condition-monitoring practice, which has typically treated impulsivity statistics and spectral or MFCC energy descriptors as interchangeable proxies for anomaly, selected on convenience rather than demonstrated cross-unit or cross-type portability. Prior one-class approaches to acoustic fault detection have generally reported feature performance within a single training unit or machine type, leaving open whether an apparent advantage reflects a property of the descriptor itself or an artifact of the specific unit sampled during training. The slider-trained result reframes this question by showing that scores flagged as target-'like' under a kurtosis or crest-factor detector transfer across slider units and into valve units with a markedly smaller observed drop than spectral or MFCC-based scoring achieves under the same protocol, a comparison that earlier single-unit or single-type studies were not structured to make. Rather than adding another case to a growing list of feature comparisons, this finding advances the literature by demonstrating that transferability itself is conditional on the source machine type: claims of a universally preferred anomaly descriptor cannot be sustained without specifying which machine class supplied the training distribution. The slider-anchored advantage therefore stands as a positive, type-specific result rather than a generic endorsement of impulsivity statistics.

3.2 Answer to the Research Question

The evidence licenses a scope-limited but confident answer rather than a blanket recommendation. When the training source is a slider unit, transient-impulsivity statistics, namely kurtosis and crest factor, generalize to other slider units and to valve units with an observed drop that is smaller than that of conventional spectral or MFCC energy features, and this pattern holds under a single-feature ablation and under a group-disjoint evaluation that removes leakage as a competing account. For the slider-trained condition specifically, impulsivity scoring is the preferable choice for a one-class detector built on slider data. This finding does not extend to the general question of which feature family a practitioner should default to without knowing the training unit's machine type, since the boundary analysis shows the same impulsivity statistics behave differently when the source distribution is a valve rather than a slider. What the study licenses, precisely, is a conditional claim: transient-impulsivity statistics are the better-generalizing descriptor under one-class training on sliders, and this preference is a property of the machine-type pairing rather than of impulsivity features as a class.

3.3 Scope and Limitations

These scope bounds follow directly from the design and should be read as limits on reach, not as reservations about what was found. The analysis rests on a small corpus of short, single-channel recordings spanning only two machine types, so the slider-favoring pattern for transient-impulsivity statistics is established within that pairing and is not licensed beyond it. The four impulsivity statistics, time-domain kurtosis, crest factor, spectral kurtosis, and envelope sparsity, serve as a proxy for the broader construct of transient impulsivity, and the ablation and leakage checks are computed on the same recordings and one-class detectors as the lead analysis, so they sharpen and stress-test a single body of evidence rather than supplying independent replications; the reported robustness and boundary results should be read as internal consistency checks on one dataset rather than as separate confirmations from separate samples. Statistical power is bounded accordingly: the study licenses claims about the direction and relative size of the generalization gap under one-class training on sliders, not claims about its magnitude across arbitrary factory settings, sensor placements, fault types, or machine populations beyond solenoid valves and slide rails. Within that explicit boundary, the finding stands as established.

4 Methods

We evaluate whether spectral or impulsivity acoustic descriptors better support cross-domain anomaly detection for industrial machine sounds, using a corpus of 1784 ten-second clips sampled at 16 kHz across two machine types (slider, valve) with four IDs each (id_00, id_02, id_04, id_06), yielding 8 groups. Of these, 787 normal clips are reserved for fitting and the remaining 997 held-out clips (880 anomaly, 117 normal) are used only for AUC scoring. Two feature families are extracted per clip: a 32-dimensional spectral/MFCC set (13 MFCC means, 13 MFCC variances, spectral centroid, spectral flatness, and four log-spaced band-energy ratios) and a 5-dimensional impulsivity set (time-domain excess kurtosis, crest factor, spectral kurtosis of frame-wise STFT energy, Hilbert-envelope Gini index, and the rate of 100 ms windows exceeding five times local RMS). For each group, a one-class detector is fit exclusively on that group's normal train clips after StandardScaler fitting on train data only; the primary detector is EllipticEnvelope with robust covariance (support fraction 0.9, contamination 0.1), with IsolationForest (200 trees, contamination 0.1) serving as an ablation and robustness check. Evaluation is strictly group-disjoint: AUC is computed for every ordered train-group by test-group cell in an 8x8 matrix per feature set and detector, yielding within-domain (diagonal) and cross-domain (off-diagonal) values from which generalization gaps are derived across 56 ordered pairs.

4.1 Feature Extraction and Detectors

The spectral/MFCC set concatenates 13 MFCC means and 13 MFCC variances with spectral centroid, spectral flatness, and four log-spaced band-energy ratios into a 32-dimensional vector, computed via librosa and scipy Welch routines. The impulsivity set is 5-dimensional and comprises time-domain excess kurtosis, crest factor defined as

$$\mathrm{CF} = \frac{\max_t |x(t)|}{\mathrm{RMS}(x)} \tag{1}$$

spectral kurtosis computed as the kurtosis of frame-wise STFT energy using a window length of 1024 and hop size of 512, the Gini index of the Hilbert-envelope amplitude, and the rate of 100 millisecond window samples whose local energy exceeds five times the local RMS. For each of the eight groups, a StandardScaler is fit exclusively on that group's normal train clips, and features from every clip are standardized before detection. The primary detector, **EllipticEnvelope**, uses robust covariance estimation with support fraction 0.9 and contamination 0.1, flagging anomalies via the robust Mahalanobis distance

$$D_M(x) = \sqrt{(x-\hat{\mu})^\top \hat{\Sigma}^{-1}(x-\hat{\mu})} \tag{2}$$

where $\hat{\mu}$ and $\hat{\Sigma}$ are the robust location and covariance estimates fit on that group's scaled normal train clips. As an ablation and robustness detector, **IsolationForest** is fit with 200 trees and contamination 0.1, scoring each clip by its average isolation path length across the ensemble.

4.2 Evaluation Protocol

The 8 groups are defined by crossing 2 machine types with 4 IDs. For each group, the StandardScaler is fit on that group's normal train clips (drawn from the 787 available across all groups) and applied unchanged to the held-out test clips (997 total: 880 anomaly, 117 normal), so that no held-out statistic contaminates the fitted scaler. From each scaled clip, the two feature sets described above are computed independently. Each detector (EllipticEnvelope and IsolationForest) is fit separately per group on that group's scaled normal train clips, using one feature set at a time, and scored on the scaled held-out test clips from every one of the 8 groups, yielding a full 8x8 matrix of train-group by test-group AUC values per feature set and detector combination. The generalization gap for each ordered pair is the within-domain (diagonal) AUC minus the cross-domain (off-diagonal) AUC for that train-group/test-group cell, producing 56 ordered pairs per feature set and detector combination that supply the inputs to the statistical comparisons below.

4.3 Statistical Analysis and Robustness Checks

The 56 ordered generalization-gap pairs per feature set and detector combination form the sample for all statistical comparisons. The headline pooled contrast is additionally assessed via a paired bootstrap with 10000 resamples drawn with replacement over the 56 pairs, recomputing the mean paired difference on each resample to obtain a bootstrap distribution and interval for the pooled effect. AUC values, entering the 8x8 matrices and hence all gaps and pairs, are computed only on the 997 held-out test clips (880 anomaly, 117 normal); the 787 normal train clips are used solely for fitting. A robustness checklist, run separately from the hypothesis tests, varied the EllipticEnvelope support fraction across 0.7, 0.75, 0.8, 0.85, 0.9, and 0.95, contamination across 0.05, 0.1, 0.15, and 0.2, and the scaler choice between standard and robust scaling.

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